

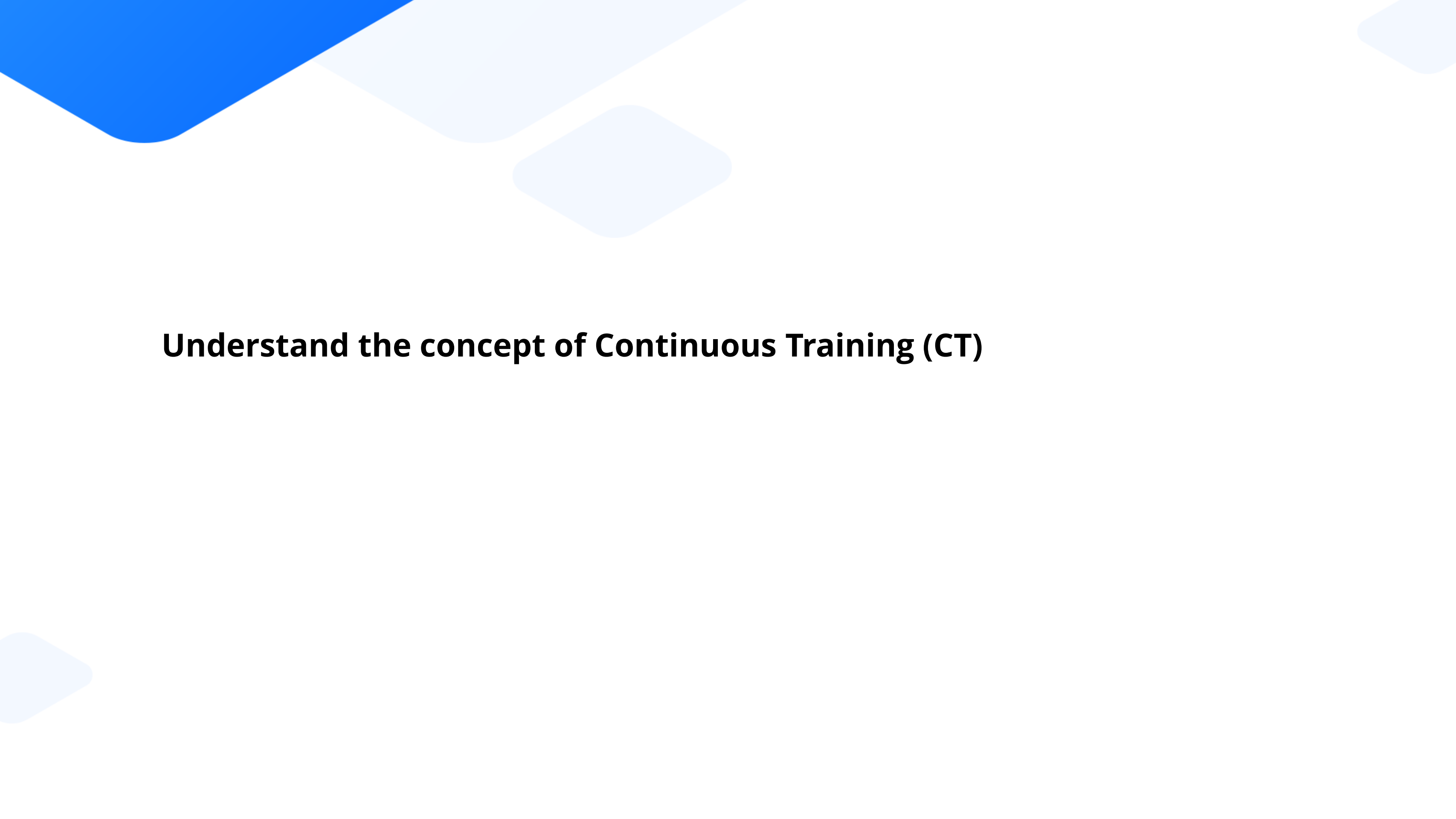
M04 - Building Continuous Training (CT) Pipelines



Topic

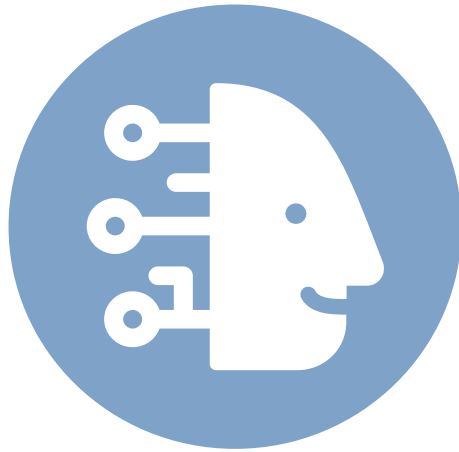
- **Understand the concept of Continuous Training (CT)**
- **Explain why AI models require periodic retraining**
- **Differentiate between event-driven and scheduled retraining**
- **Understand how training pipelines are orchestrated**
- **Learn the basics of pipeline automation**
- **Apply best practices for Continuous Training**
- **Build a simple automated retraining pipeline**





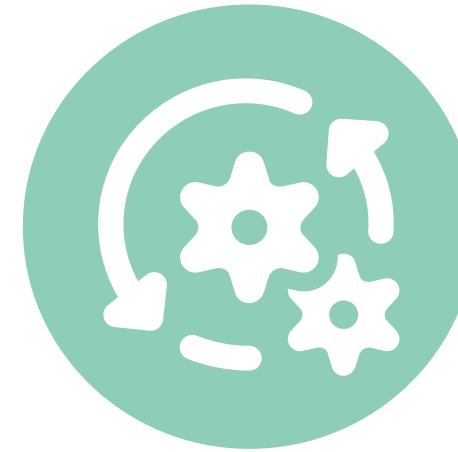
Understand the concept of Continuous Training (CT)

Choose the best approach for AI model training.



Manual Training

Requires developer intervention for retraining



Continuous Training

Automates retraining for ongoing accuracy

Continuous Training (CT) Overview

Reduces manual effort

Automating data preparation, model training, and validation.

Ensures reliable predictions

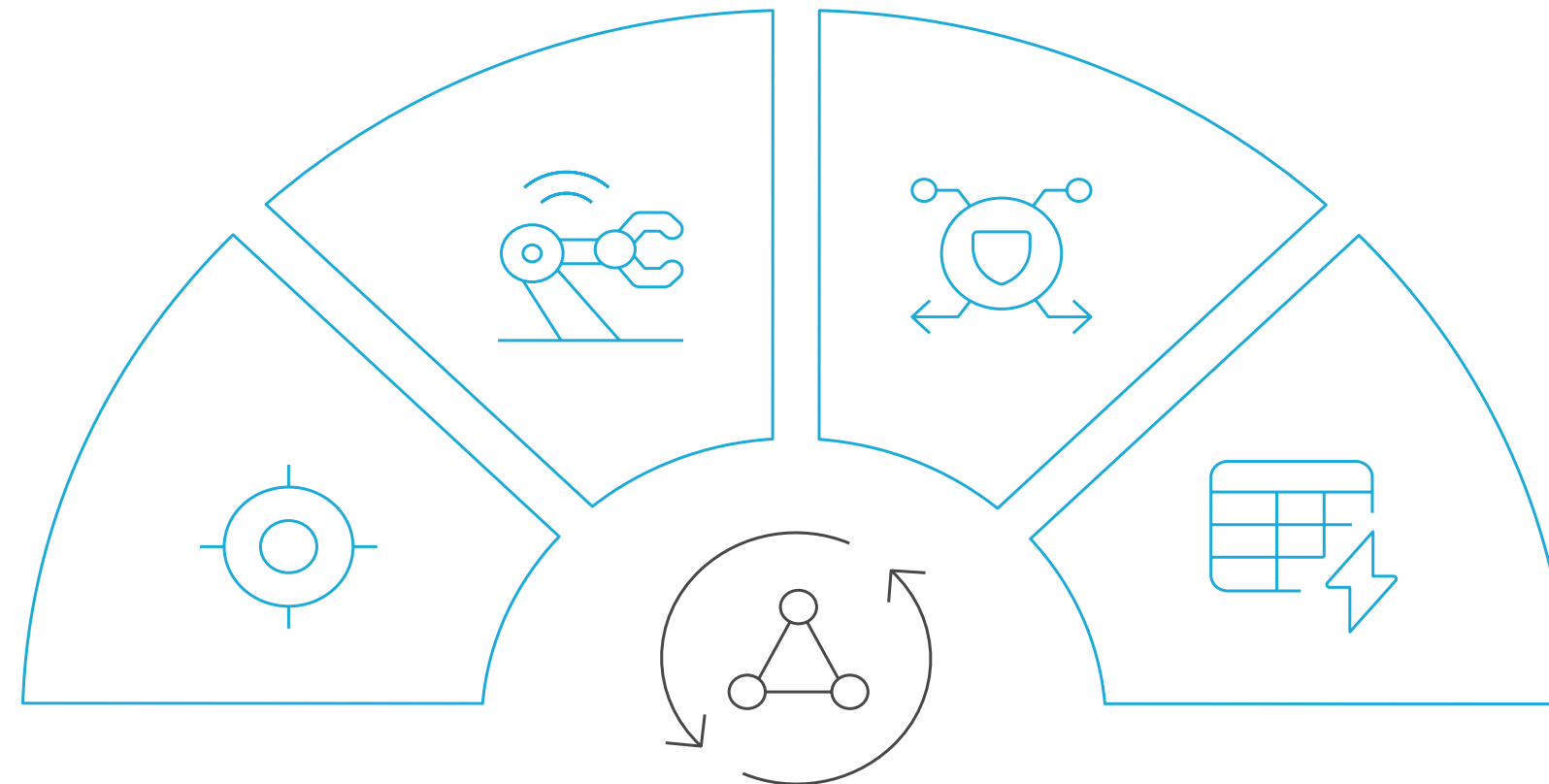
Maintaining reliability as applications evolve with every new release.

Keeps models accurate

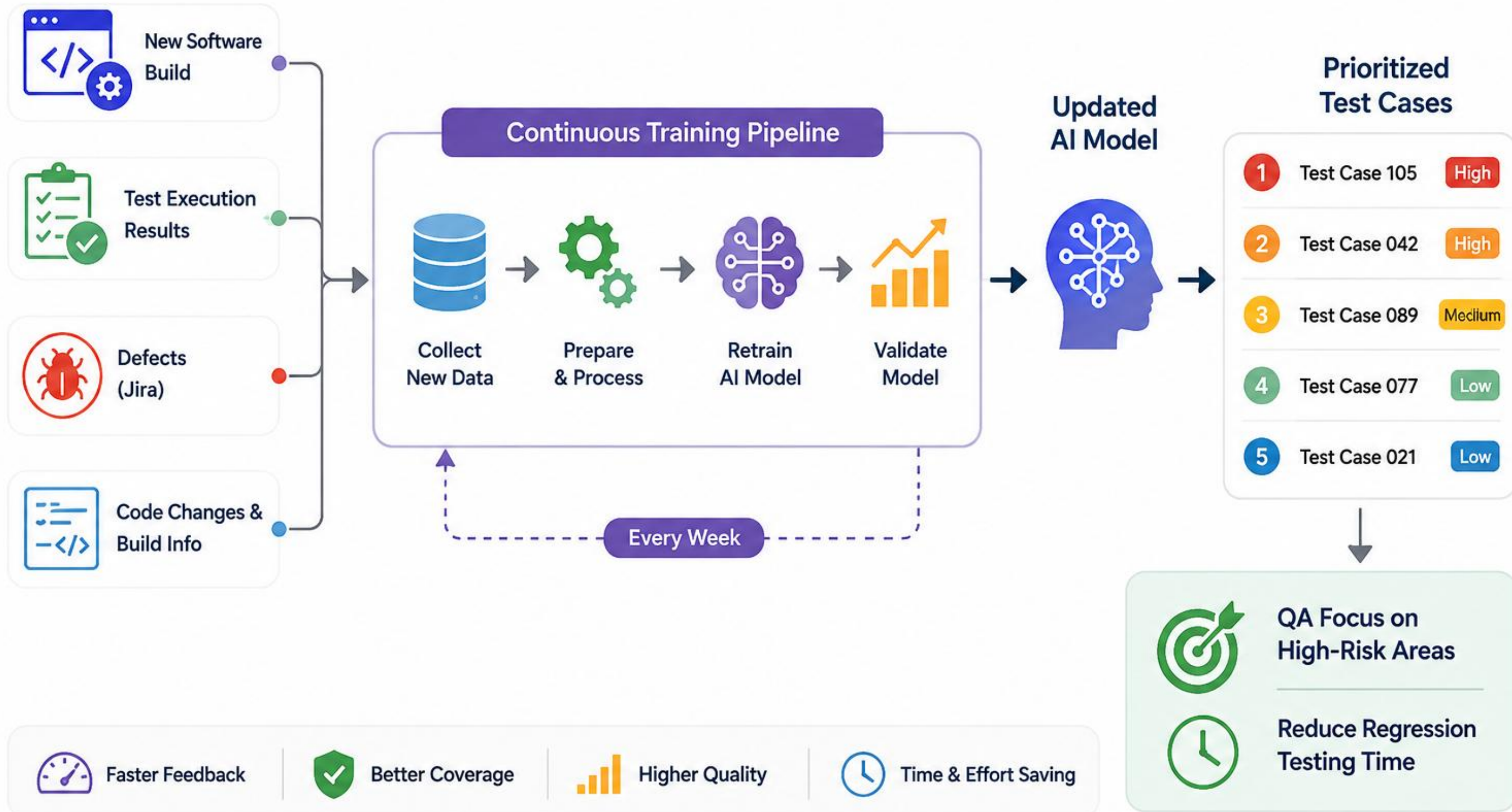
Learning from the latest test execution results and software changes.

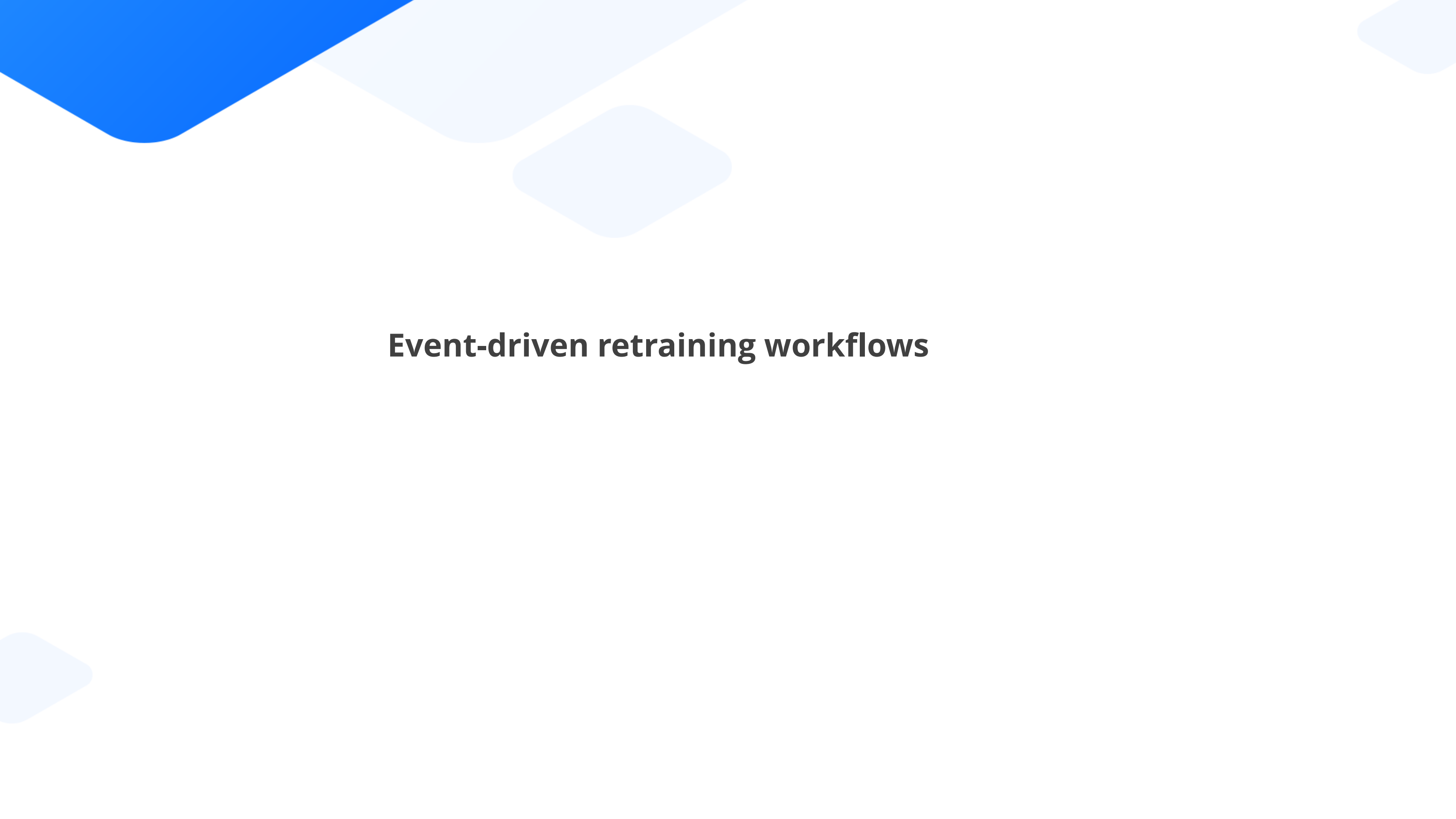
Supports faster testing

Continuously improving AI-based recommendations for smarter testing.



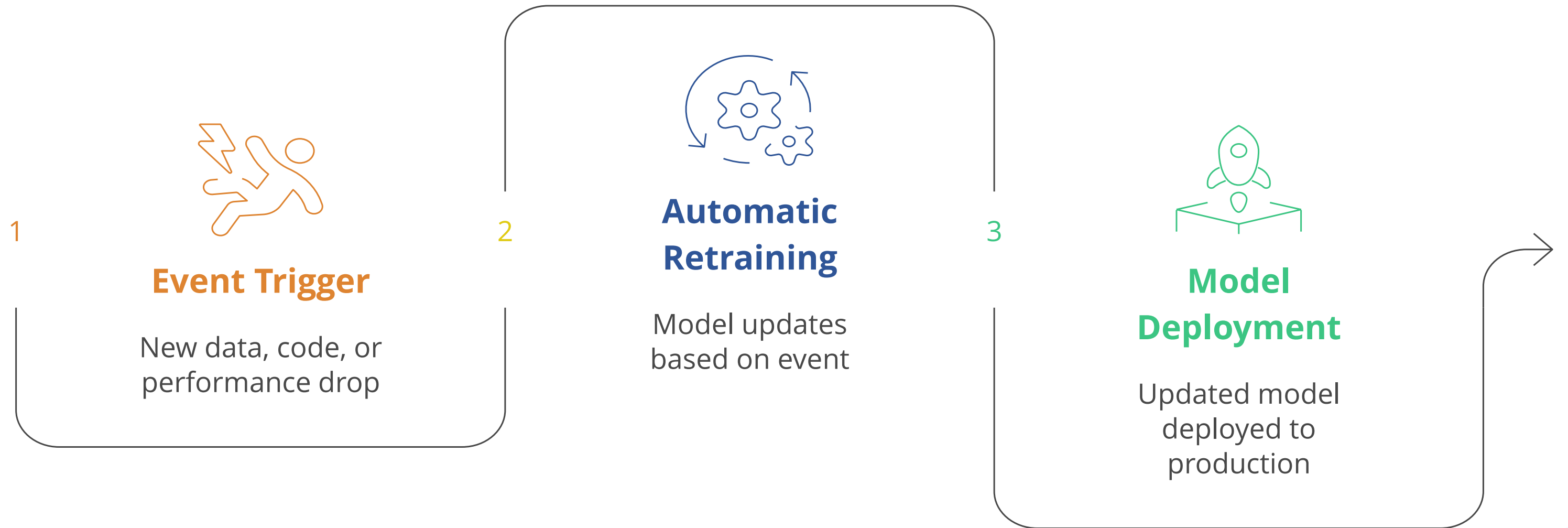
VOIS Example – AI-Based Test Case Prioritization





Event-driven retraining workflows

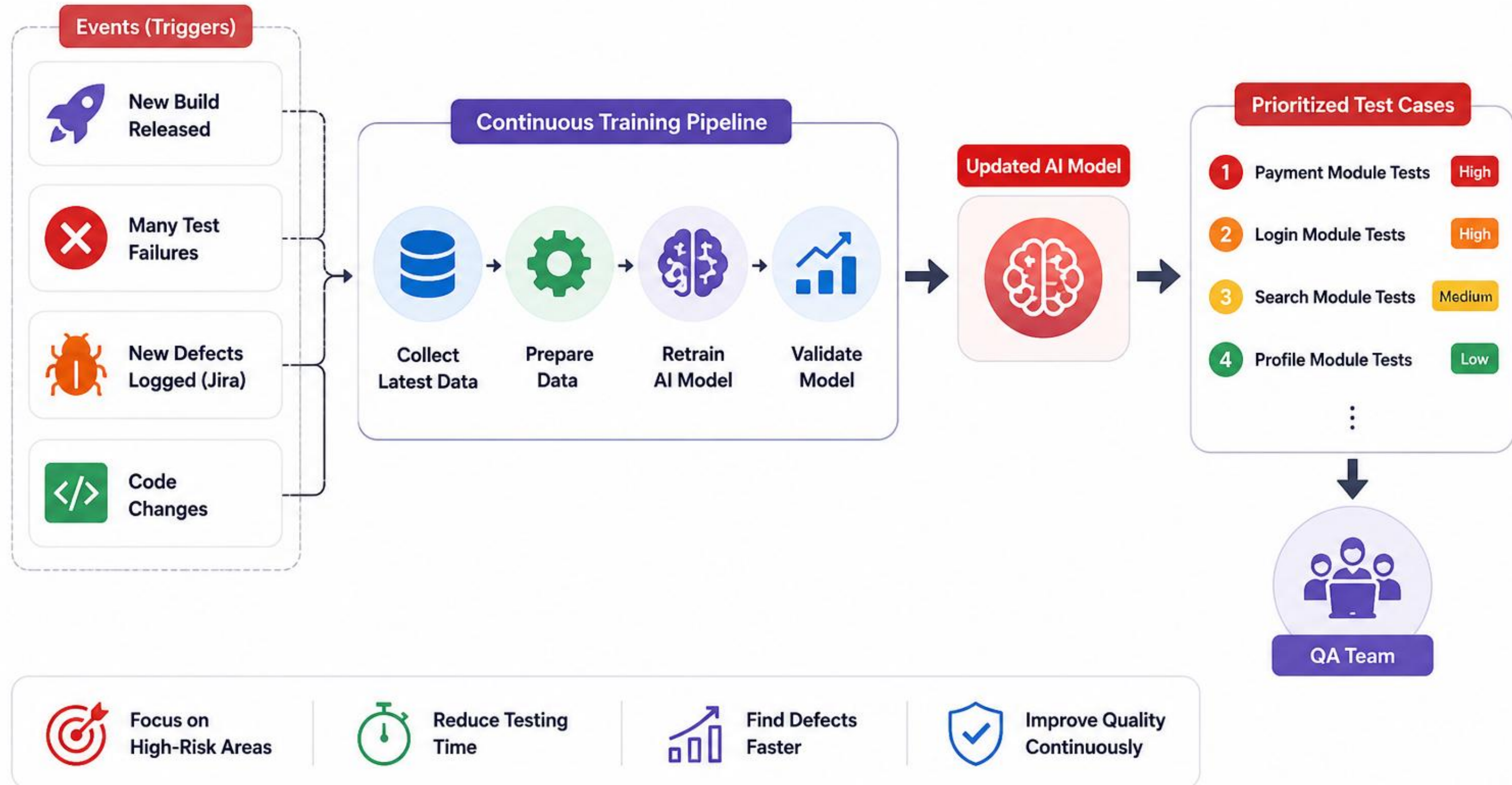
Event-driven retraining workflows





Event-Driven Retraining Workflow

AI-Based Test Case Prioritization

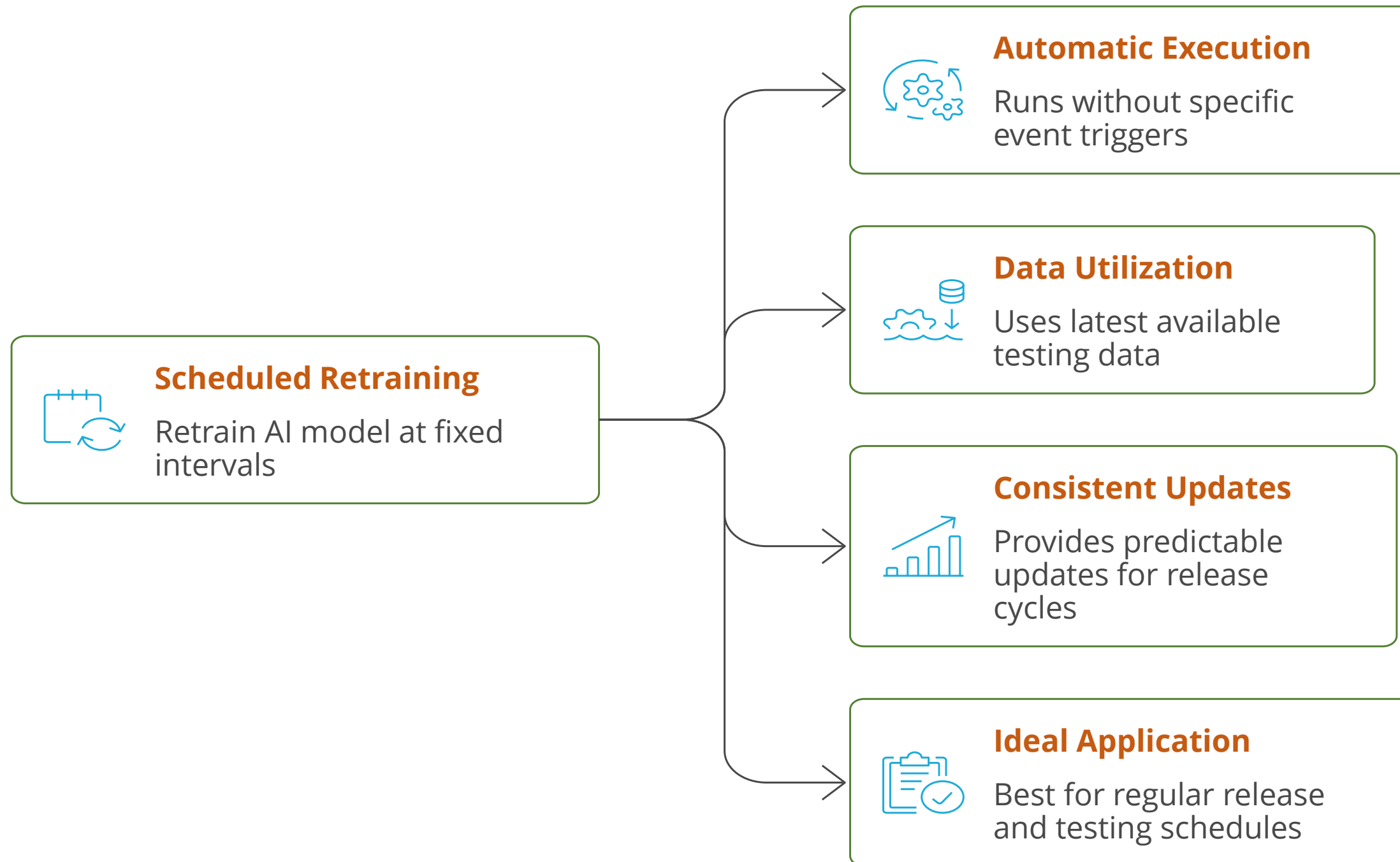




Scheduled Retraining Approaches

Scheduled Retraining Approaches

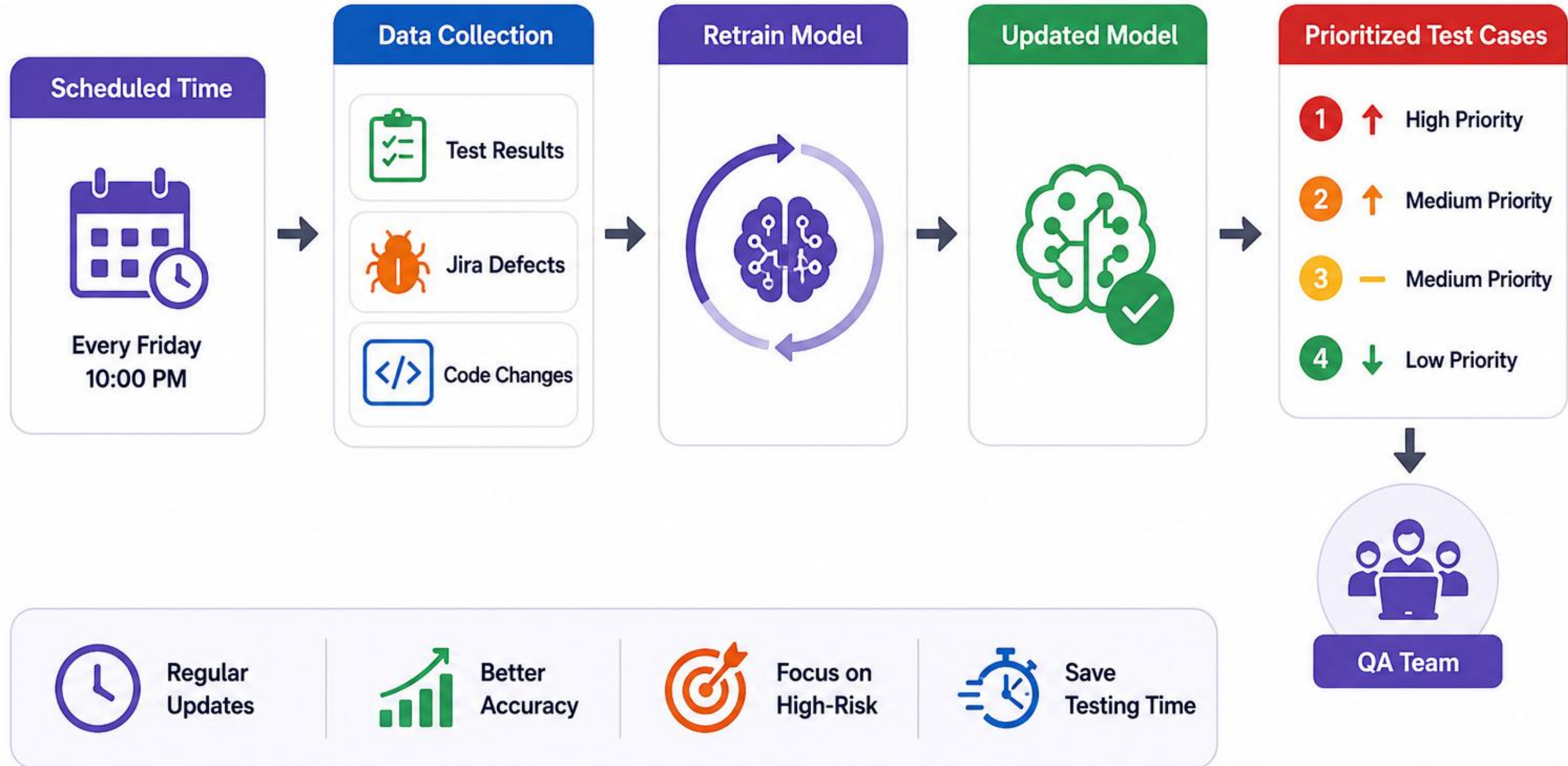
Scheduled Retraining means the AI model is retrained automatically at a fixed time (for example, every Friday), even if no special event has occurred.

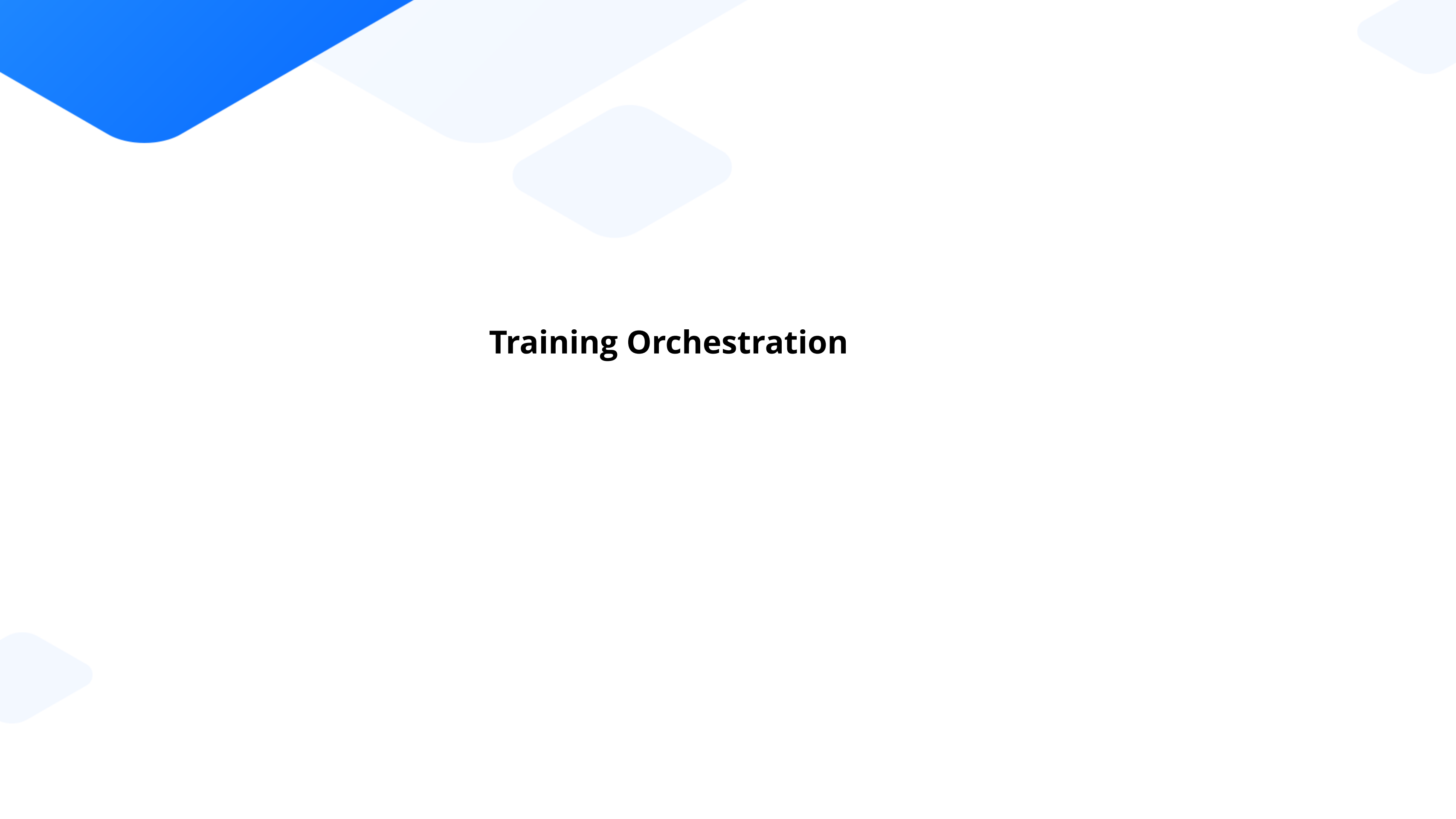




Scheduled Retraining Approach

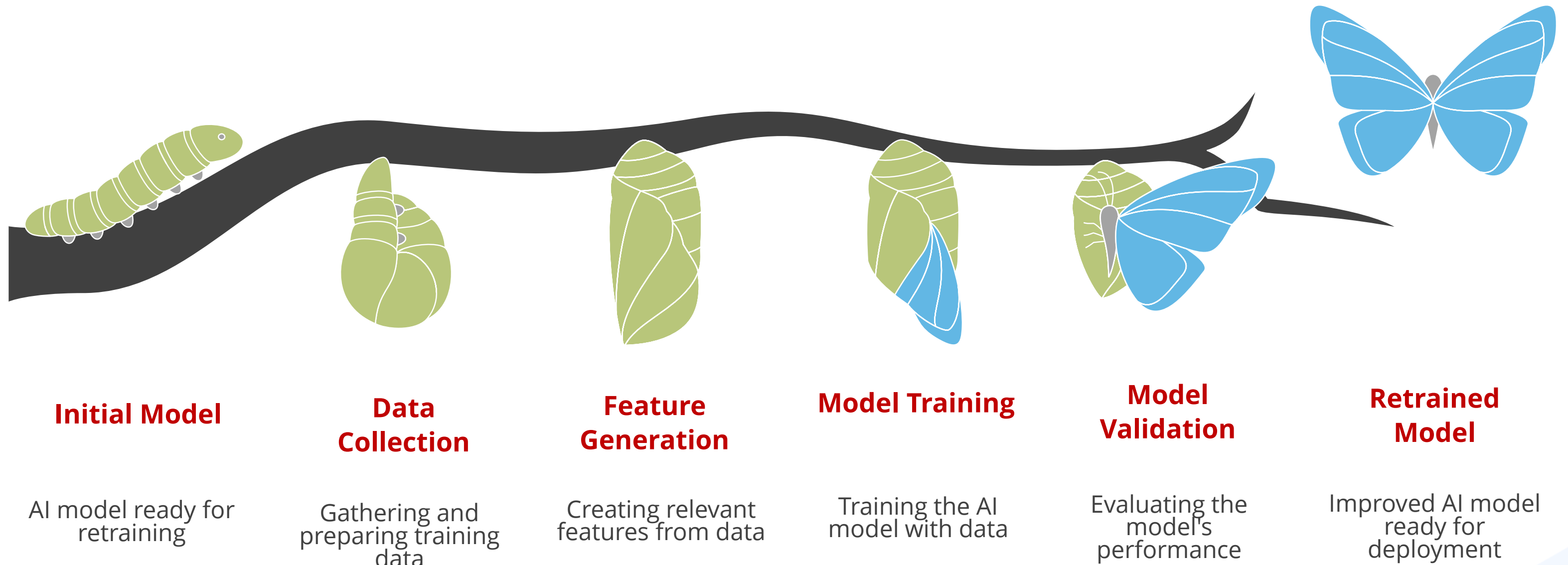
AI-Based Test Case Prioritization





Training Orchestration

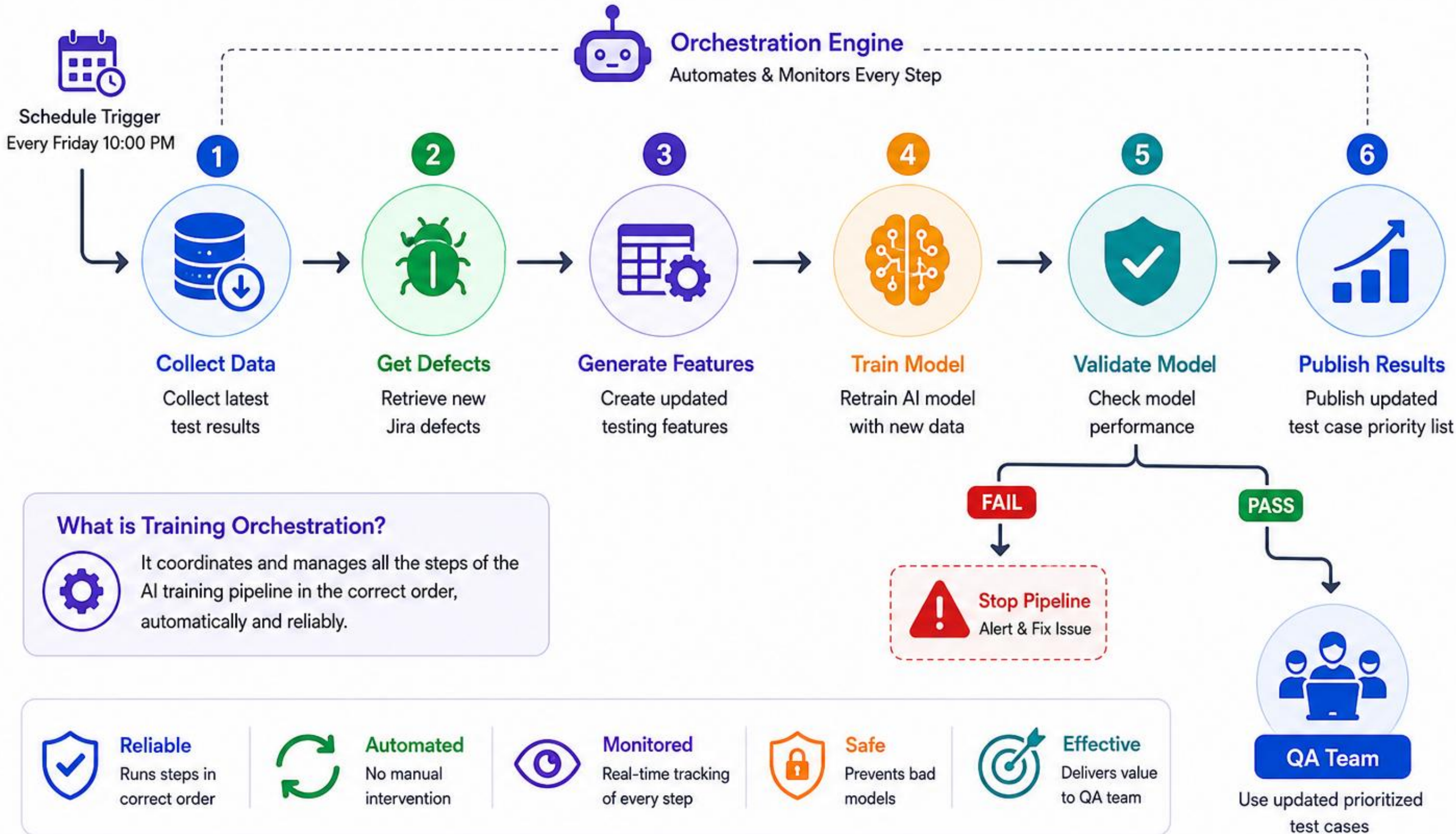
Training Orchestration





Training Orchestration

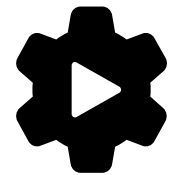
Orchestrating Every Step of Continuous Training





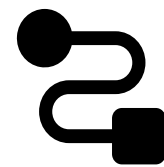
Pipeline automation concepts

Pipeline automation concepts



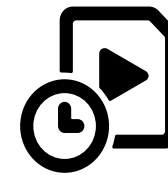
**Execute AI
workflow**

Executes the complete AI workflow automatically with minimal manual effort.



**Run sequence
steps**

Each step runs in sequence, from collecting data to generating predictions.



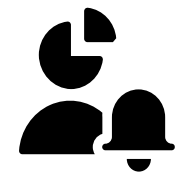
**Reduce
manual work**

Reduces manual work and speeds up the AI model update process.



**Ensure
process
consistency**

Ensures every run follows the same process, improving consistency and reliability.



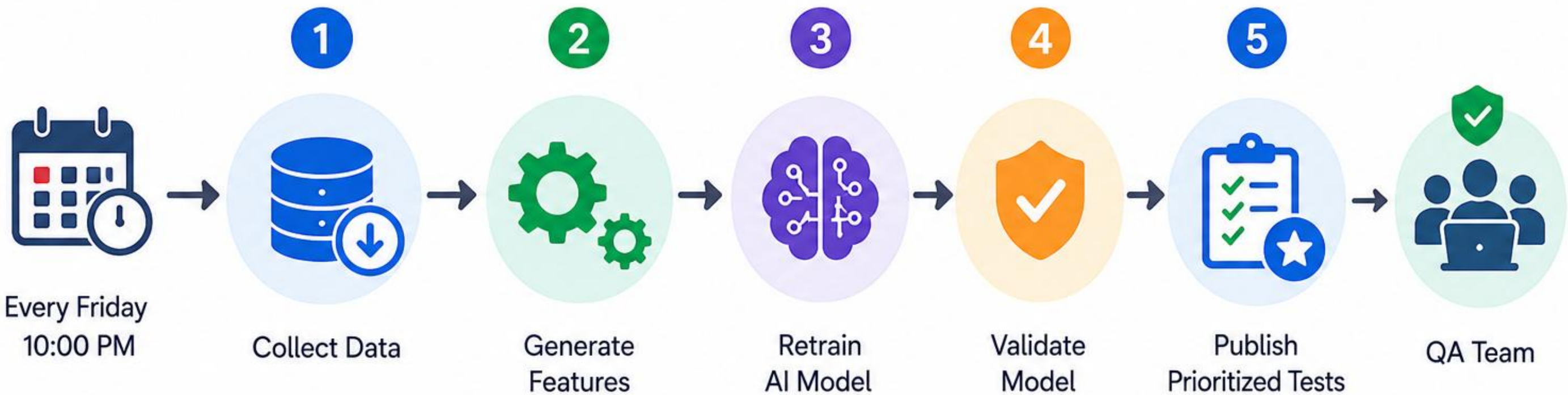
**Report
pipeline status**

Automatically reports success or failure, making it easier to monitor the pipeline.



Pipeline Automation Concepts

Automate • Orchestrate • Deliver Value



Faster Updates



Automated Workflow



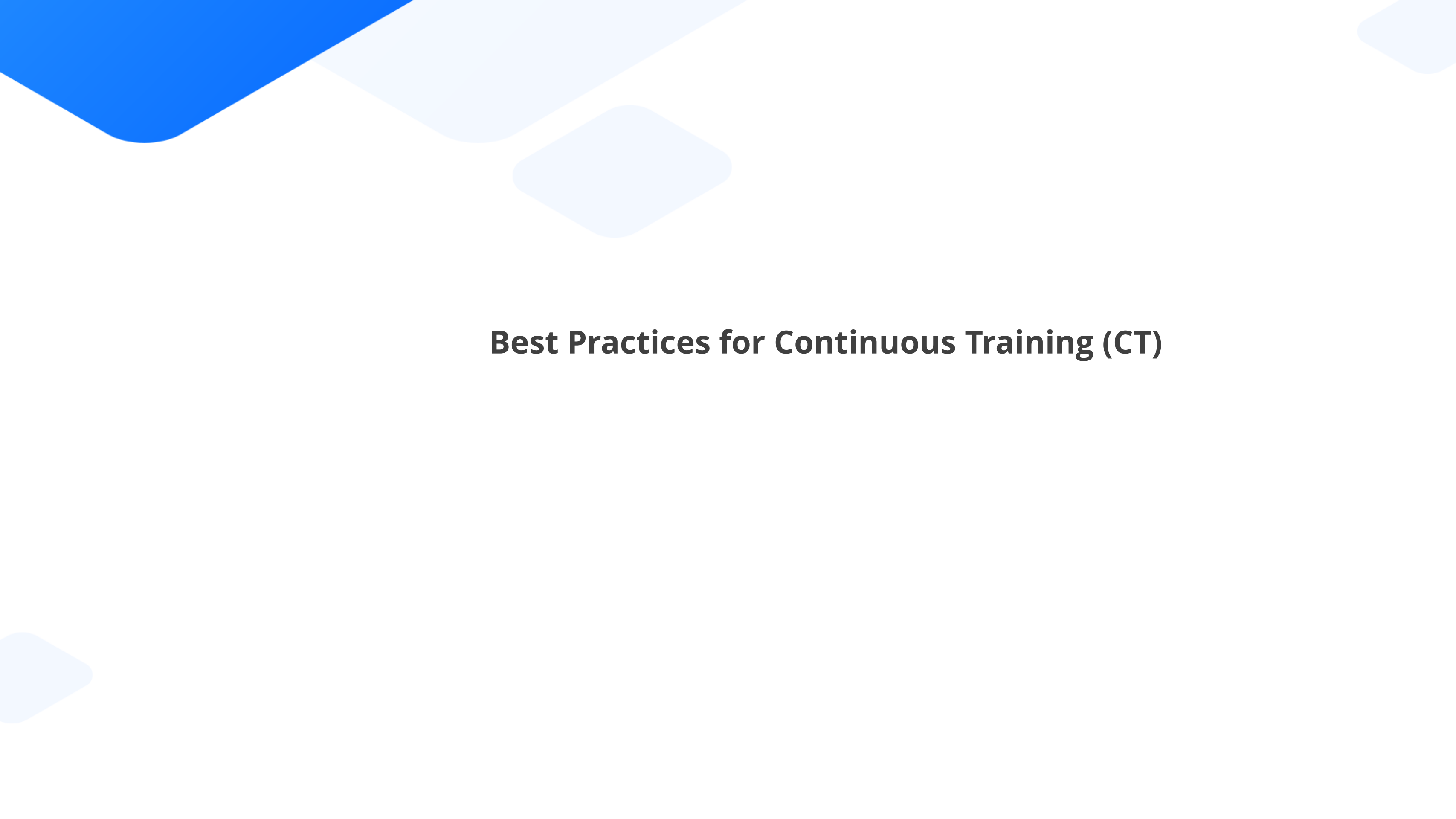
Consistent Process



Reliable Results

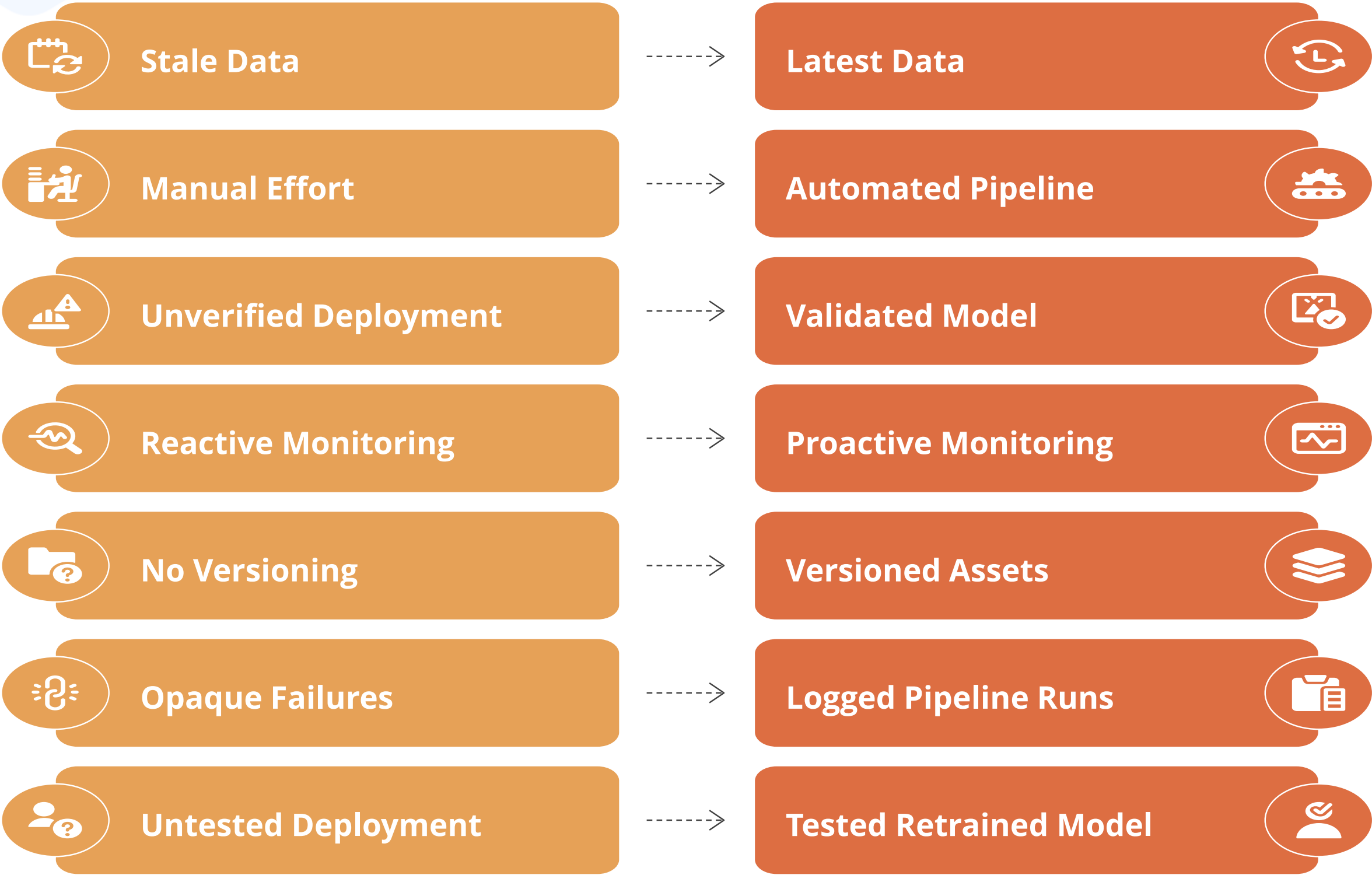


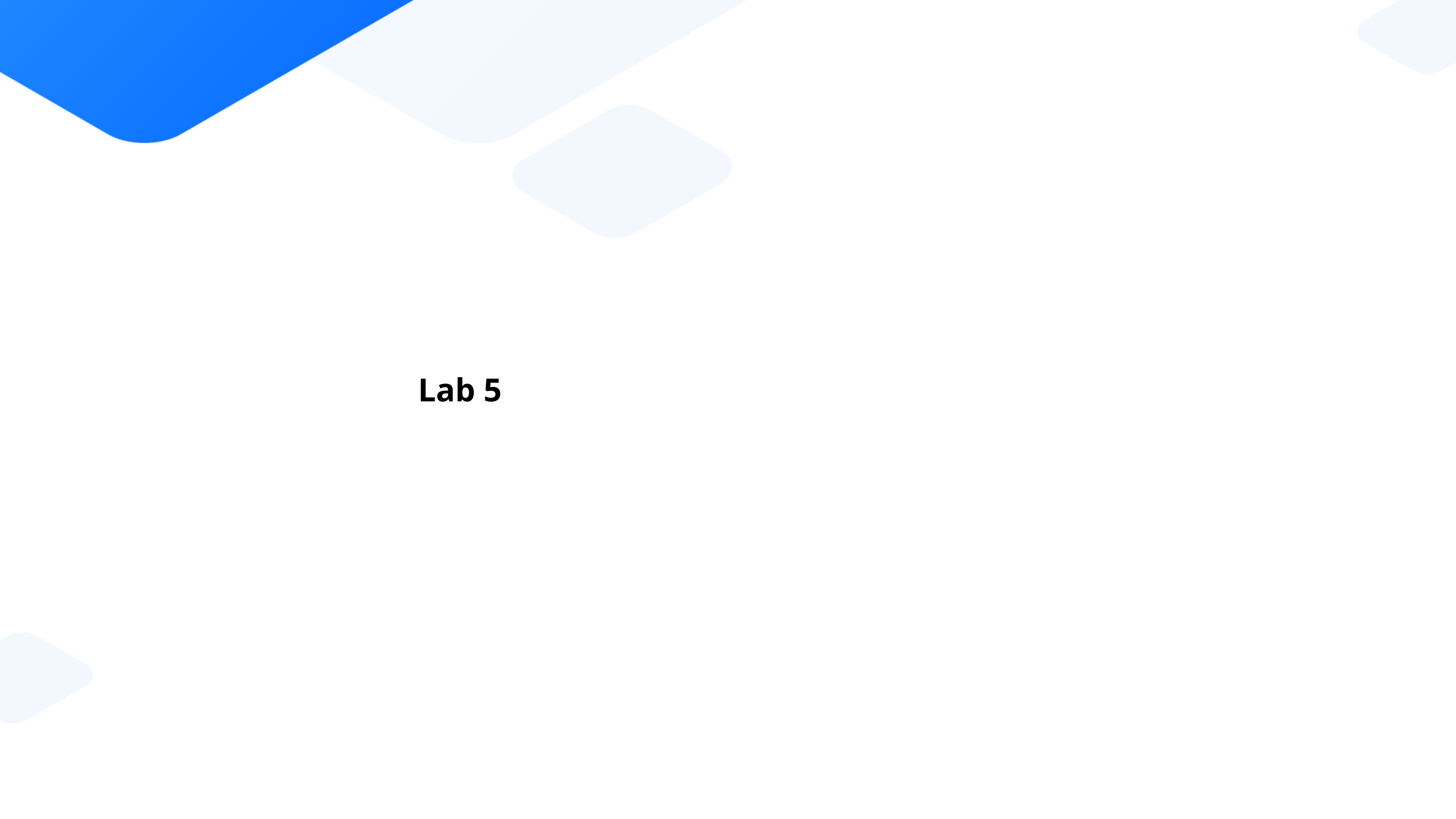
Easy Monitoring



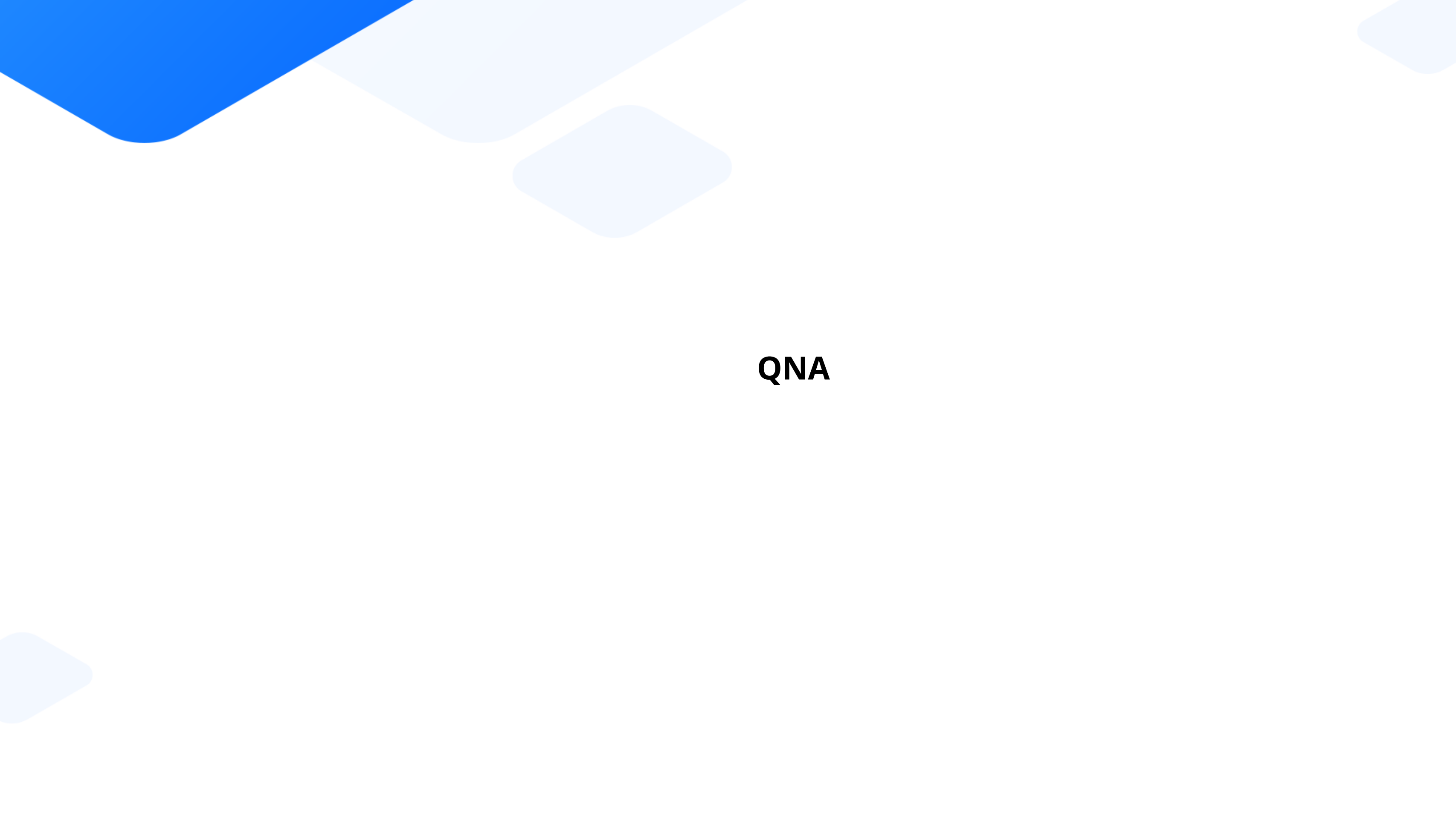
Best Practices for Continuous Training (CT)

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Lab 5



QNA